

以下是从您提供的四篇文章中提取的5个段落（每段大约110-120词），并附上了对应的中文翻译：

段落一：选自《吴恩达来信：如何成为多语言编程者》

English:

If you move from one programming language to another that carries out similar tasks but with different syntax - say, from JS to TS, or C++ to Java, or Rust to Go once you've learned the first set of concepts, you'll know a lot of the concepts needed to prompt an LLM to code in the second language. (Although TensorFlow and PyTorch are not programming languages, learning the concepts of deep learning behind TensorFlow will also make it much easier to get an LLM to write PyTorch code for you, and vice versa!) In addition, you'll be able to understand much of the generated code (perhaps with a little LLM assistance).

中文翻译:

如果你从一种编程语言转向另一种执行相似任务但语法不同的语言——比如，从JS转向TS，或者从C++转向Java，或者从Rust转向Go——一旦你学习了第一套概念，你就会了解很多提示大型语言模型（LLM）用第二种语言编写代码所需的概念。（虽然TensorFlow和PyTorch不是编程语言，但学习TensorFlow背后的深度学习概念也会让你更容易让LLM为你编写PyTorch代码，反之亦然！）此外，你将能够理解大部分生成的代码（也许需要LLM的一点协助）。

段落二：选自《人工智能正在进化，像人类一样处理世界》

English:

Most artificial intelligence systems are primarily created on the foundation of deep learning algorithms. This is the means of exposing a computer program to thousands of examples and AI learning how to solve problems through this process. Deep learning can be boiled down to teaching a computer how to be smart. After any given deep learning phase for an AI, the system can perceive the inputs that it was trained on and make decisions therein. The decision-making tree that the AI forms from traditional deep learning mimics the way the right side of our brain works. It is based on the perception of inputs, of pseudo-senses.

中文翻译:

大多数人工智能系统主要是建立在深度学习算法的基础上的。这是一种让计算机程序接触数千个示例，并让人工智能通过这个过程学习如何解决问题的方法。深度学习可以归结为教计算机如何变聪明。在人工智能经历了任何特定的深度学习阶段之后，系统就能感知其接受过训练的输入内容并在其中做出决策。人工智能从传统的深度学习中形成的决策树，模仿了我们右脑的工作方式。它是基于对输入的感知，即对伪感官的感知。

段落三：选自《让计算机理解人类的情感》

English:

Lead researcher Philip Kragel, who works as a research associate at the University of Colorado Boulder's Institute of Cognitive Science, fed the neural network 25,000 images and asked the system to sort them into 20 categories of emotions, some subtle. The list included emotions like anxiety and boredom, but also less obvious human emotional experiences like "aesthetic appreciation" and "empathic pain." Analyzing the images, the neural network made sense of what it saw by parsing the facial expressions and body posture of humans depicted. Then it was time to see how Emonet's emotion categorization skills compared to that of the average human brain.

中文翻译:

科罗拉多大学博尔德分校认知科学研究所的副研究员、首席研究员菲利普·克拉格尔（Philip Kragel）向该神经网络输入了2.5万张图像，并要求系统将它们分为20种情绪类别，其中一些情绪非常微妙。该列表包括焦虑和无聊等情绪，但也包括不太明显的人类情感体验，如“审美欣赏”和“共情痛苦”。在分析这些图像时，神经网络通过解析所描绘的人类的面部表情和身体姿势来理解它所看到的内容。接下来，就该看看EmoNet的情绪分类技能与普通人类大脑相比如何了。

段落四：选自《计算机代码究竟是语言还是数学？》

English:

The study, carried out by a team from MIT and Tufts University, had two dozen participants read code, English, and logic puzzles while in an fMRI machine. By seeing which parts of the brain lit up while doing these tasks, the researchers could determine how our brains process coding languages. If the areas of the brain associated with language processing were to light up, then we treat code like we treat languages. The

same would go for the math parts. The control tasks, reading either a real sentence or a nonsense one and memorizing the location of colored squares, demonstrated the baseline activation levels for these systems in each subject.

中文翻译:

这项由麻省理工学院和塔夫茨大学团队开展的研究，让二十多名参与者在功能磁共振成像（fMRI）机器中阅读代码、英语和逻辑谜题。通过观察在执行这些任务时大脑的哪些部分会亮起，研究人员就可以确定我们的大脑是如何处理编程语言的。如果与语言处理相关的大脑区域亮起，那就说明我们像对待语言一样对待代码。数学部分也是如此。对照任务，如阅读一个真实的句子或一个无意义的句子，以及记住彩色方块的位置，展示了每个受试者大脑中这些系统的基线激活水平。

段落五：选自《计算机代码究竟是语言还是数学？》

English:

When the subjects were in the machine, they were asked to work through the code and predict the output. The brain scans showed only limited responses in the brain's language processing centers, but a considerable amount in the multiple demand (MD) system, which often handles math, logic, and executive tasks. While this may sound like a win for the "coding is math" argument, it isn't quite the slam dunk you might think it is. This system handles most of our "difficult" thinking and is useful for many things. Logic and math typically cause the left half of it to fire up while the right half handles abstract thinking.

中文翻译:

当受试者在机器中时，他们被要求通读代码并预测输出结果。脑部扫描显示，大脑的语言处理中心的反应非常有限，但在多重需求（MD）系统中有相当大的反应，该系统通常处理数学、逻辑和执行任务。虽然这听起来像是“编码就是数学”这一论点的胜利，但这并不像你想象的那么确凿。这个系统处理了我们大部分的“困难”思考，并在很多方面都很有用。逻辑和数学通常会激活它的左半球，而右半球则处理抽象思维。